



Configuring RIPv2 to Implement Dynamic Routing

Difficulty (out of 5 stars): ★★★

Recommended study time: 2 hours

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1 About This Lab

1.1 Task Description

Routing Information Protocol (RIP) is a simple Interior Gateway Protocol (IGP). RIP was designed for small and medium-sized networks that use the same technology.

RIP employs the hop count to measure the distance to a destination. The distance is called the metric value. In RIP, the default hop count from a routing device to its directly connected network is 0, and the hop count from a routing device to a network that is reachable through another routing device is 1. Each additional routing device increases the hop count by one. That is, the total hop count (metric value) equals the number of routing devices along the path from the local network to the destination network. To restrict the convergence time, RIP requires that the metric value be an integer ranging from 0 to 15. A metric value greater than or equal to 16 is defined as infinite, which means that the destination network or host is unreachable. As such, RIP is not suitable for large-scale networks.

After completing this task, you will be able to independently set up and configure small networks using RIP.

1.2 Objectives

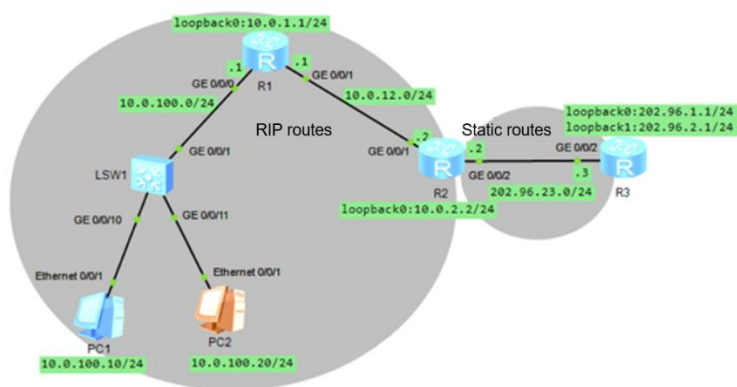
On completing this task, you will be able to:

1. Configure RIP.
2. Understand the function of a silent RIP interface.
3. Configure RIP to advertise default routes.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology



2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
```



```
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int lo0
[R1-LoopBack0]ip address 10.0.1.1 24
[R1-LoopBack0]quit
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.0.100.1 24
[R1-GigabitEthernet0/0/0]quit
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.0.12.1 24
[R1-GigabitEthernet0/0/1]quit
[R1]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int lo0
[R2-LoopBack0]ip address 10.0.2.2 24
[R2-LoopBack0]quit
[R2]int g0/0/1
[R2-GigabitEthernet0/0/1]ip address 10.0.12.2 24
[R2-GigabitEthernet0/0/1]quit
[R2]int g0/0/2
[R2-GigabitEthernet0/0/2]ip address 202.96.23.2 24
[R2-GigabitEthernet0/0/2]quit
[R2]
```

- Initial configuration of R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]int lo0
[R3-LoopBack0]ip address 202.96.1.1 24
[R3-LoopBack0]quit
[R3]int lo1
[R3-LoopBack1]ip address 202.96.2.1 24
[R3-LoopBack1]quit
[R3]int g0/0/2
[R3-GigabitEthernet0/0/2]ip address 202.96.23.3 24
[R3-GigabitEthernet0/0/2]quit
[R3]
[R3]ip route-static 10.0.0.0 255.255.0.0 202.96.23.2
[R3]
```



- The initial configurations of PC1 and PC2 are shown in Figure 2 and Figure 3.

Figure 2-2 IP address of PC1

```
PC1
基础配置 命令行 组播 UDP发包工具 串口
Welcome to use PC Simulator!

PC>ipconfig

Link local IPv6 address.....: fe80::5689:98ff:fe4f:7ea0
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.0.100.10
Subnet mask.....: 255.255.255.0
Gateway.....: 10.0.100.1
Physical address.....: 54-89-98-4F-7E-A0
DNS server.....:

PC>ping 10.0.100.1

Ping 10.0.100.1: 32 data bytes, Press Ctrl_C to break
From 10.0.100.1: bytes=32 seq=1 ttl=255 time=63 ms
From 10.0.100.1: bytes=32 seq=2 ttl=255 time=46 ms
From 10.0.100.1: bytes=32 seq=3 ttl=255 time=47 ms
From 10.0.100.1: bytes=32 seq=4 ttl=255 time=32 ms
From 10.0.100.1: bytes=32 seq=5 ttl=255 time=46 ms

--- 10.0.100.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 32/46/63 ms

PC>
```



Figure 2-3 IP address of PC2

```
PC2
 基础配置  命令行  组播  UDP发包工具  串口
Welcome to use PC Simulator!

PC>ipconfig

Link local IPv6 address.....: fe80::5689:98ff:feeb:56bb
IPv6 address.....: :: / 128
IPv6 gateway.....: ::
IPv4 address.....: 10.0.100.20
Subnet mask.....: 255.255.255.0
Gateway.....: 10.0.100.1
Physical address.....: 54-89-98-EB-56-BB
DNS server.....:

PC>ping 10.0.100.1

Ping 10.0.100.1: 32 data bytes, Press Ctrl_C to break
From 10.0.100.1: bytes=32 seq=1 ttl=255 time=62 ms
From 10.0.100.1: bytes=32 seq=2 ttl=255 time=31 ms
From 10.0.100.1: bytes=32 seq=3 ttl=255 time=32 ms
From 10.0.100.1: bytes=32 seq=4 ttl=255 time=47 ms
From 10.0.100.1: bytes=32 seq=5 ttl=255 time=31 ms

--- 10.0.100.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/40/62 ms

PC>
```

3 Task Implementation

3.1 Checking Basic Configurations

Run the **display ip interface brief** command on R1, R2, and R3 to check interface configurations.

```
<R1>display ip int brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
```

```
The number of interface that is UP in Physical is 4
The number of interface that is DOWN in Physical is 1
The number of interface that is UP in Protocol is 4
The number of interface that is DOWN in Protocol is 1
```

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	10.0.100.1/24	up	up
GigabitEthernet0/0/1	10.0.12.1/24	up	up
GigabitEthernet0/0/2	unassigned	down	down
LoopBack0	10.0.1.1/24	up	up(s)
NULL0	unassigned	up	up(s)

```
<R1>
```



```
<R2>display ip int brief
```

```
*down: administratively down
```

```
^down: standby
```

```
(l): loopback
```

```
(s): spoofing
```

```
The number of interface that is UP in Physical is 4
```

```
The number of interface that is DOWN in Physical is 1
```

```
The number of interface that is UP in Protocol is 4
```

```
The number of interface that is DOWN in Protocol is 1
```

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	down	down
GigabitEthernet0/0/1	10.0.12.2/24	up	up
GigabitEthernet0/0/2	202.96.23.2/24	up	up
LoopBack0	10.0.2.2/24	up	up(s)
NULL0	unassigned	up	up(s)

```
<R2>
```

```
<R3>display ip int brief
```

```
*down: administratively down
```

```
^down: standby
```

```
(l): loopback
```

```
(s): spoofing
```

```
The number of interface that is UP in Physical is 4
```

```
The number of interface that is DOWN in Physical is 2
```

```
The number of interface that is UP in Protocol is 4
```

```
The number of interface that is DOWN in Protocol is 2
```

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	down	down
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	202.96.23.3/24	up	up
LoopBack0	202.96.1.1/24	up	up(s)
LoopBack1	202.96.2.1/24	up	up(s)
NULL0	unassigned	up	up(s)

```
<R3>
```

Run the **ping** command to check the connectivity between R1 and other devices.

```
<R1>ping 10.0.100.10
```

```
PING 10.0.100.10: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.100.10: bytes=56 Sequence=1 ttl=128 time=80 ms
```

```
Reply from 10.0.100.10: bytes=56 Sequence=2 ttl=128 time=90 ms
```

```
Reply from 10.0.100.10: bytes=56 Sequence=3 ttl=128 time=100 ms
```

```
Reply from 10.0.100.10: bytes=56 Sequence=4 ttl=128 time=90 ms
```

```
Reply from 10.0.100.10: bytes=56 Sequence=5 ttl=128 time=80 ms
```

```
--- 10.0.100.10 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```



```
round-trip min/avg/max = 80/88/100 ms
```

```
<R1>ping 10.0.12.2
PING 10.0.12.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=100 ms
  Reply from 10.0.12.2: bytes=56 Sequence=2 ttl=255 time=80 ms
  Reply from 10.0.12.2: bytes=56 Sequence=3 ttl=255 time=70 ms
  Reply from 10.0.12.2: bytes=56 Sequence=4 ttl=255 time=70 ms
  Reply from 10.0.12.2: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 10.0.12.2 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 30/70/100 ms
<R1>
```

Run the **ping** command to check the connectivity between R2 and other devices.

```
<R2>ping 202.96.23.3
PING 202.96.23.3: 56 data bytes, press CTRL_C to break
  Reply from 202.96.23.3: bytes=56 Sequence=1 ttl=255 time=50 ms
  Reply from 202.96.23.3: bytes=56 Sequence=2 ttl=255 time=20 ms
  Reply from 202.96.23.3: bytes=56 Sequence=3 ttl=255 time=30 ms
  Reply from 202.96.23.3: bytes=56 Sequence=4 ttl=255 time=20 ms
  Reply from 202.96.23.3: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 202.96.23.3 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 20/30/50 ms

<R2>
```

Run the **display ip routing-table** command to check the static routes configured on R3.

```
<R3>display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 14          Routes : 14

Destination/Mask    Proto    Pre  Cost           Flags NextHop         Interface
-----
   10.0.0.0/16      Static   60    0           RD   202.96.23.2 GigabitEthernet 0/0/2
   127.0.0.0/8       Direct   0     0           D    127.0.0.1      InLoopBack0
   127.0.0.1/32      Direct   0     0           D    127.0.0.1      InLoopBack0
  127.255.255.255/32 Direct   0     0           D    127.0.0.1      InLoopBack0
   202.96.1.0/24     Direct   0     0           D    202.96.1.1     LoopBack0
   202.96.1.1/32     Direct   0     0           D    127.0.0.1      LoopBack0
   202.96.1.255/32   Direct   0     0           D    127.0.0.1      LoopBack0
```



```

202.96.2.0/24 Direct 0 0 D 202.96.2.1 LoopBack1
202.96.2.1/32 Direct 0 0 D 127.0.0.1 LoopBack1
202.96.2.255/32 Direct 0 0 D 127.0.0.1 LoopBack1
202.96.23.0/24 Direct 0 0 D 202.96.23.3 GigabitEthernet 0/0/2
202.96.23.3/32 Direct 0 0 D 127.0.0.1 GigabitEthernet 0/0/2
202.96.23.255/32 Direct 0 0 D 127.0.0.1 GigabitEthernet 0/0/2
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
<R3>

```

3.2 Configuring RIP on R1

RIP runs only on the interfaces of a specified network segment, and does not send, receive, or forward routes on any other interfaces.

```

[R1]rip
[R1-rip-1]version 2
[R1-rip-1]network 10.0.0.0
[R1-rip-1]quit
[R1]

```

3.3 Configuring RIP on R2

```

[R2]rip
[R2-rip-1]version 2
[R2-rip-1]network 10.0.0.0
[R2-rip-1]quit
[R2]

```

3.4 Verifying RIP Routes

Check the RIP routes in the routing tables of R1 and R2.

```

[R1]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
    Destinations : 14          Routes : 14

Destination/Mask    Proto    Pre  Cost    Flags NextHop          Interface
-----
10.0.1.0/24        Direct  0    0        D   10.0.1.1      LoopBack0
10.0.1.1/32        Direct  0    0        D   127.0.0.1     LoopBack0
10.0.1.255/32      Direct  0    0        D   127.0.0.1     LoopBack0
10.0.2.0/24      RIP      100  1        D   10.0.12.2     GigabitEthernet0/0/1
10.0.12.0/24       Direct  0    0        D   10.0.12.1     GigabitEthernet 0/0/1
10.0.12.1/32       Direct  0    0        D   127.0.0.1     GigabitEthernet 0/0/1
10.0.12.255/32     Direct  0    0        D   127.0.0.1     GigabitEthernet 0/0/1
10.0.100.0/24      Direct  0    0        D   10.0.100.1    GigabitEthernet 0/0/0
10.0.100.1/32      Direct  0    0        D   127.0.0.1     GigabitEthernet 0/0/0
10.0.100.255/32    Direct  0    0        D   127.0.0.1     GigabitEthernet 0/0/0
127.0.0.0/8        Direct  0    0        D   127.0.0.1     InLoopBack0

```




```

127.0.0.1/32 Direct 0 0 D 127.0.0.1 InLoopBack0
127.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0

[R2]display ip routing-table
Route Flags: R - relay, D - download to fib
-----
Routing Tables: Public
      Destinations : 15          Routes : 15

Destination/Mask    Proto    Pre  Cost           Flags NextHop          Interface
-----
10.0.1.0/24        RIP      100  1             D    10.0.12.1 GigabitEthernet 0/0/1
10.0.2.0/24        Direct   0    0             D    10.0.2.2 LoopBack0
10.0.2.2/32        Direct   0    0             D    127.0.0.1 LoopBack0
10.0.2.255/32      Direct   0    0             D    127.0.0.1 LoopBack0
10.0.12.0/24       Direct   0    0             D    10.0.12.2 GigabitEthernet 0/0/1
10.0.12.2/32       Direct   0    0             D    127.0.0.1 GigabitEthernet 0/0/1
10.0.12.255/32     Direct   0    0             D    127.0.0.1 GigabitEthernet 0/0/1
10.0.100.0/24      RIP      100  1             D    10.0.12.1 GigabitEthernet 0/0/1
127.0.0.0/8        Direct   0    0             D    127.0.0.1 InLoopBack0
127.0.0.1/32       Direct   0    0             D    127.0.0.1 InLoopBack0
127.255.255.255/32 Direct   0    0             D    127.0.0.1 InLoopBack0
202.96.23.0/24     Direct   0    0             D    202.96.23.2 GigabitEthernet 0/0/2
202.96.23.2/32     Direct   0    0             D    127.0.0.1 GigabitEthernet 0/0/2
202.96.23.255/32   Direct   0    0             D    127.0.0.1 GigabitEthernet 0/0/2
255.255.255.255/32 Direct   0    0             D    127.0.0.1 InLoopBack0

[R2]

```

The routing tables of R1 and R2 each contain a RIP route. The value of the **Protocol** field is **RIP**, indicating that the route is a RIP route.

3.5 Verifying Network Connectivity

Ping R2's loopback0 interface (10.0.2.2) from PC1.

```

PC>ping 10.0.2.2

Ping 10.0.2.2: 32 data bytes, Press Ctrl_C to break
From 10.0.2.2: bytes=32 seq=1 ttl=254 time=62 ms
From 10.0.2.2: bytes=32 seq=2 ttl=254 time=31 ms
From 10.0.2.2: bytes=32 seq=3 ttl=254 time=47 ms
From 10.0.2.2: bytes=32 seq=4 ttl=254 time=47 ms
From 10.0.2.2: bytes=32 seq=5 ttl=254 time=47 ms

--- 10.0.2.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 31/46/62 ms

```



```
PC>
```

Run the **tracert** command to check the data transmission path.

```
PC>tracert 10.0.2.2
```

```
traceroute to 10.0.2.2, 8 hops max
(ICMP), press Ctrl+C to stop
 1  10.0.100.1    32 ms  46 ms  47 ms
 2  10.0.2.2     47 ms  47 ms  31 ms
```

```
PC>
```

3.6 Configuring Silent RIP Interfaces

Silent RIP interfaces only receive incoming packets for updating the routing table but do not send RIP packets. R1's G0/0/0 interface is connected to a terminal. Therefore, you can configure G0/0/0 as a silent interface on R1. Then, R1 does not send RIP messages to the terminal network through this interface.

```
[R1]rip
[R1-rip-1]silent-interface g0/0/0
[R1-rip-1]
[R1-rip-1]display this

[V200R003C00]
#
rip 1
 version 2
 network 10.0.0.0
 silent-interface GigabitEthernet0/0/0
#
return
[R1-rip-1]quit
[R1]
```

3.7 Configuring a Default Route to Implement Network Connectivity

On R2, configure a default route to R3.

```
[R2]ip route-static 0.0.0.0 0.0.0.0 202.96.23.3
[R2]
```

3.8 Verifying Connectivity

Verify the connectivity of the network between the Loopback0 interfaces on R2 and R3.

```
[R2]ping 202.96.1.1
PING 202.96.1.1: 56 data bytes, press CTRL_C to break
 Reply from 202.96.1.1: bytes=56 Sequence=1 ttl=255 time=30 ms
 Reply from 202.96.1.1: bytes=56 Sequence=2 ttl=255 time=40 ms
 Reply from 202.96.1.1: bytes=56 Sequence=3 ttl=255 time=20 ms
 Reply from 202.96.1.1: bytes=56 Sequence=4 ttl=255 time=20 ms
```



```
Reply from 202.96.1.1: bytes=56 Sequence=5 ttl=255 time=20 ms
```

```
--- 202.96.1.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 20/26/40 ms
```

R2 has been configured with a default route, and R3 has been configured with a static route. Therefore, R2 can ping R3.

Verify the connectivity of the network between the Loopback0 interfaces on R1 and R3.

```
<R1>ping 202.96.1.1
PING 202.96.1.1: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 202.96.1.1 ping statistics ---
 5 packet(s) transmitted
 0 packet(s) received
100.00% packet loss
```

As shown in the figure, R1 cannot ping the loopback0 interface of R3 because R1 does not have a route to the loopback0 interface.

3.9 Configuring RIP to Advertise Default Routes

By default, RIP does not advertise default routes to neighbors. On real-world networks, you can configure RIP to advertise default routes as required in order to control routing information.

Configure RIP to advertise the default route in the RIP process on R2.

```
[R2]rip
[R2-rip-1]default-route originate cost 2
[R2-rip-1]quit
[R2]
```

Check the RIP routes on R1.

```
[R1]display ip routing-table protocol rip
Route Flags: R - relay, D - download to fib

-----
Public routing table : RIP
      Destinations : 2          Routes : 2

RIP routing table status : <Active>
      Destinations : 2          Routes : 2
```

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	RIP	100	3	D	10.0.12.2	GigabitEthernet 0/0/1



```
10.0.2.0/24  RIP    100  1          D  10.0.12.2  GigabitEthernet 0/0/1

RIP routing table status : <Inactive>
    Destinations : 0          Routes : 0

[R1]
```

The command output shows that the R1 routing table contains a default RIP route.

3.10 Verifying Network Connectivity

On R1, ping the IP address 202.96.1.1 of the loopback interface on R3 again.

```
[R1]ping 202.96.1.1
  PING 202.96.1.1: 56  data bytes, press CTRL_C to break
    Reply from 202.96.1.1: bytes=56 Sequence=1 ttl=254 time=20 ms
    Reply from 202.96.1.1: bytes=56 Sequence=2 ttl=254 time=30 ms
    Reply from 202.96.1.1: bytes=56 Sequence=3 ttl=254 time=30 ms
    Reply from 202.96.1.1: bytes=56 Sequence=4 ttl=254 time=30 ms
    Reply from 202.96.1.1: bytes=56 Sequence=5 ttl=254 time=30 ms

--- 202.96.1.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/28/30 ms

[R1]
```

On PC1, ping the IP address 202.96.1.1 of the loopback interface on R3.

```
PC>ping 202.96.1.1

Ping 202.96.1.1: 32 data bytes, Press Ctrl_C to break
From 202.96.1.1: bytes=32 seq=1 ttl=253 time=63 ms
From 202.96.1.1: bytes=32 seq=2 ttl=253 time=46 ms
From 202.96.1.1: bytes=32 seq=3 ttl=253 time=32 ms
From 202.96.1.1: bytes=32 seq=4 ttl=253 time=31 ms
From 202.96.1.1: bytes=32 seq=5 ttl=253 time=47 ms

--- 202.96.1.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 31/43/63 ms

PC>
```

Run the **tracert** command on PC1 to check the packet forwarding path.

```
PC>tracert 202.96.1.1

tracert to 202.96.1.1, 8 hops max
(ICMP), press Ctrl+C to stop
```



```
1 10.0.100.1 31 ms 47 ms 47 ms
2 10.0.12.2 31 ms 47 ms 47 ms
3 202.96.1.1 47 ms 62 ms 63 ms
```

PC>

The command output shows that the packets are forwarded by R1 (10.0.100.1) to R2 (10.0.12.2), and then to R3 (202.96.1.1).

4 Verification

Verification has been performed during the lab.

5 Quiz

1. What is the function of a silent RIP interface?
2. In the lab, if no default route to R3 is configured on R2, can PC1 ping R3?
3. What are the application scenarios of RIP?